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Configurational Sampling of All-Atom Solvated Membranes Using Hybrid Nonequilibrium Molecular Dynamics Monte Carlo Simulations

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Abstract

All-atom simulations are a powerful approach to study the structure and dynamics of biological membranes. However, sampling the atomic configurations of inhomogeneous membranes can be challenging due to the slow lateral diffusion of the various constituents. To address this issue, a hybrid nonequilibrium molecular dynamics - Monte Carlo (neMD/MC) simulation method is proposed in which randomly chosen lipid molecules are swapped to generate configurations that are subsequently accepted or rejected according to a Metropolis criterion based on the alchemical work for the attempted swap calculated via a short trajectory. A dual topology framework constraining the common atoms of the exchanging molecules yields a good acceptance

probability using switching trajectories as short as 10 ps. The performance of the hybrid neMD/MC algorithm and its ability to sample the distribution of lipids near a transmembrane helix carrying a net charge are illustrated for a binary mixture of charged and zwitterionic lipids.

Keywords: Alchemical transformation, Monte–Carlo, Metropolis criteria, molecular dynamics simulation, nonequilibrium dynamics, membrane.

Membranes are essential components of living organisms. They consist of amphiphilic lipid molecules organized in a bilayer hosting a variety of intrinsic membrane proteins involving a wide range of biological functions.[1] The lipidome of living organism comprises a considerable number of chemically different lipids.[2] They assemble in various proportions to give the biological membranes specific properties that vary according to the species, cells of a given organism or organelles.[3,4] It is now well-established that lipid composition affects not only the mechanical properties of membranes but also modulates the function of membrane proteins through distinct mechanisms.[5,6] Despite recent advances in experimental studies of lipid-protein interaction, including high-resolution structures capturing lipid binding sites, our understanding of the spatial organization of membranes around proteins remains limited.[7]

Molecular simulations emerged as an attractive approach to provide atomic-level information regarding both the structure and dynamics of biological membranes. All-atom molecular dynamics (MD) are often employed to study the structure and dynamics of membrane proteins in simple homogeneous phospholipid bilayers. It is however clear that the accessible simulation timescales are not sufficient to allow a satisfactory sampling of inhomogeneous multi-component membranes where lipid diffusion is very slow.[8,9] For this reason, many simulation studies of lipid-protein association in complex mixtures are based on coarse-grained (CG) MD simulations, in which groups of atoms are reduced to a single effective particle.[10,11] The speed-up of the calculations puts the study of complex membrane dynamics

within reach, yet at the cost of an approximate representation of the molecular system.[7,9]

Very few methods aimed at sampling the configuration space of inhomogeneous bilayer using all-atom simulations have been proposed in recent years. Some of them relies on generalized-ensemble algorithms such as replica-exchange molecular dynamics (REMD).[12,13] Replicas running at higher temperature allows for a faster diffusion of the lipids, resulting in an enhanced sampling of bilayer configurations. Another category consists in driving the system via short nonequilibrium MD (neMD) trajectories to generate a new state of the system that are subsequently accepted or rejected via a Metropolis MC step.[14–17] These hybrid simulation methods combining the advantages of MC with the strengths of MD offer promising strategies to efficiently sample the configurations of complex molecular systems such as membranes. The concept of the neMD/MC algorithm was first introduced in the context of constant-pH simulations[18–20] whereby the protons are coupled and decoupled alchemically from the rest of the system, leading to sampling of all protonation states according to their proper Boltzmann weight. More generally, the ability of neMD/MC simulations to sample equilibrium configurations provide an important tool for studying complex biomolecule systems.[21–24]

Kindt and coworkers reported the first example of a hybrid MD/MC simulation of a bilayer involving lipid mutations.[25–27] In their work, mutation was limited to lipids differing by a very few atoms. Every accepted mutation modified the bilayer composition thereby, generating configurations in the grand canonical ensemble. More recently, Fathizadeh and Elber developed the MDAS algorithm (Molecular Dynamics with Alchemical Steps) aimed at sampling efficiently multi-component fluids by means of MC exchange between pair of molecules selected randomly on the basis of alchemical work calculated from a nonequilibrium MD trajectory.[28] They assessed the efficiency of MDAS by studying the mixing of a binary lipid system made of POPC and DOPC, two lipids differing by one unsaturation and a slightly longer acyl chain. The nonequilibrium work associated with the alchemical swapping was evaluated through a single-topology paradigm, involving a single copy of all the common

atoms of DOPC and POPC supplemented by the atoms belonging solely to each of these two lipids. They assessed further the efficiency by studying the phase transition in a membrane composed of DPPC and DLPC, two lipids differing by the acyl chain length. They concluded that the efficiency of such neMD/MC hybrid approaches depends crucially on the mutation strategy.[29] The MDAS algorithm has been extended to sample mixing of coarse-grained (CG) lipid mixtures.[30] Cherniavskyi et al. showed that the exchange rate for lipids carrying different charges was high in the context of the CG Martini force field. Yet, attempts exchanges of the same lipids described in all-atom models were essentially rejected. While the previous work with hybrid neMD/MC approaches provided great advances, there remains unresolved challenges in the context of all-atom simulations.

Here, we designed a general hybrid neMD/MC method aimed at sampling detailed atomic models of biological membranes. Specifically, we propose a strategy based on a dual-topology paradigm for swapping lipids that differ in their molecular structure. The method is illustrated and its performance critically evaluated in the context of a binary mixture of both zwitterionic and negatively charged lipids, DLPC and DLPG respectively. We also demonstrate the potential of our approach towards sampling the specific association of lipids to a transmembrane peptide bearing negative charges.

Method

The objective is to sample configurations according to equilibrium Boltzmann statistics. The hybrid scheme is based on a series of equilibrium MD simulations interspersed with discrete stochastic transitions in which one randomly picks two chemically different lipid molecules, a and b and attempts to exchange their position according to a Monte Carlo (MC) algorithm. If the attempted exchange is rejected, another exchange is proposed. After a series of rejected exchanges, the equilibrium propagation can be resumed without disturbance, whereas the equilibrium propagation is continued from the new position if the attempted swap is accepted.

The exchange process follows a hybrid neMD/MC scheme designed to allow configurational transition between two states.[14–16,18]

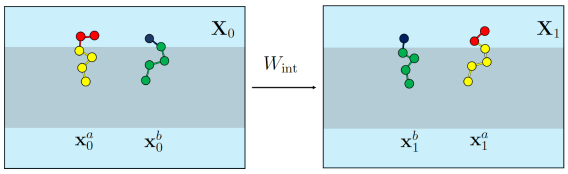


Figure 1: Schematic representation of a possible lipid exchange move. $\mathbf{x}_0^a$ and $\mathbf{x}_0^b$ are the coordinates and velocities of the lipids a and b at their initial sites and $\mathbf{x}_1^a$ and $\mathbf{x}_1^b$ in their final state after the exchange. $\mathbf{X}_o$ and $\mathbf{X}_1$ are the coordinates and velocities of the rest of the system. Red and blue spheres represent the non common atoms of the exchanging lipids, while yellow and green spheres stand for their common part. W_{int} is the nonequilibrium work associated to the lipid exchange.

The goal of the exchange process is depicted schematically in Figure 1. Briefly, it is assumed that the two molecules have a core of identical chemical topology, for which atoms can be matched one-for-one. The coordinates of the molecules a and b are written as $\mathbf{r}^a = (\overline{\mathbf{r}}^a, \underline{\mathbf{r}}^a)$ and $\mathbf{r}^b = (\overline{\mathbf{r}}^b, \underline{\mathbf{r}}^b)$, respectively, where $\overline{\mathbf{r}}^a$ and $\overline{\mathbf{r}}^b$ represent the coordinates of the atoms of the identical core, while $\underline{\mathbf{r}}^a$ and $\underline{\mathbf{r}}^b$ represent the coordinates of the remaining "dangling" atoms. The number of dangling atoms in the two molecules is not necessarily the same. We denote the remaining coordinates of the surrounding environment as $\mathbf{R}$. Conceptually, the entire swapping process consists in decoupling molecules a and b from their surrounding environment, exchanging the atomic coordinates of their common core, generating proper equilibrium coordinates for their dangling atoms, and re-coupling the exchanged molecules to their surrounding environment. The nonequilibrium work, W_{int}, associated with the complete process is then used in a Metropolis MC to accept or reject the proposed exchange.

For the sake of clarity, we write the positions and velocities of molecule a, b and the surrounding environment as, $\mathbf{x}^a \equiv (\mathbf{r}^a, \mathbf{v}^a)$, $\mathbf{x}^b \equiv (\mathbf{r}^b, \mathbf{v}^b)$, and $\mathbf{X} \equiv (\mathbf{R}, \mathbf{V})$, respectively. The exchange transition begins at $(\mathbf{x}_0^a, \mathbf{x}_0^b, \mathbf{X}_0)$ and ends at $(\mathbf{x}_1^a, \mathbf{x}_1^b, \mathbf{X}_1)$. During the alchemical nonequilibrium simulation carried out within the dual topology framework of NAMD, the two original copies a and b are progressively switched off (selected as "outgoing" in the NAMD alchemical free energy syntax) as the coupling parameter λ varies from 0 to 1, while two new copies b' and a' are simultaneously switched on (selected as "incoming" in the alchemical

free energy NAMD syntax). The complete transition $(\mathbf{x}_0^a, \mathbf{x}_0^b, \mathbf{X}_0 \rightarrow \mathbf{x}_1^a, \mathbf{x}_1^b, \mathbf{X}_1)$ is carried out in a step-wise fashion. Preliminary tests involving exchanges of lipids truncated to their polar head-groups in bulk water and exchanges of entire lipids in a membrane pointed out the need to resort to a dual topology strategy to reduce the statistical noise associated with the calculation of out-of-equilibrium work computed along a neMD/MC simulation. Employing the latter strategy becomes particularly crucial for estimating the work associated to exchanges in highly crowded molecular environment such as lipid bilayers.

The overall algorithm employed to evaluate the nonequilibrium work associated to a lipid exchange is illustrated in Figure 2. In a first step, a copy of molecules a and b is created in vacuum, referred to as a' and b'. This step aims at creating the copies of a and b in a conformation and position as close as possible to b and a, respectively, in order to limit the steric hindrance when inserted in the membrane. Therefore, the copies a' and b' have exchanged coordinates for the common core atoms, $\mathbf{r}_0^{a'} = (\overline{\mathbf{r}}_0^b, \underline{\mathbf{r}}_0^{a'})$ and $\mathbf{r}_0^{b'} = (\overline{\mathbf{r}}_0^a, \underline{\mathbf{r}}_0^{b'})$. The positions of the non-common dangling atoms of molecules a' and b' are initially constructed consistently with the position of the common core atoms in their new conformation, and are subsequently equilibrated via a simulation in vacuum. For this step in vacuum, all the atoms of molecules a and b as well as the common core atoms of a' and b' are fixed to their initial positions. An additional weak distance restraint between one dangling atom of a (or b) and one atom of b' (or a') is imposed to favor representative conformations in which the positions of the dangling atoms of molecules a' and b' overlap with those of the molecules b and a, respectively. The computationally inexpensive vacuum simulation is carried out as long as necessary to generate a conformation of a' and b' consistent with the equilibrium distribution of a dual topology system. Contributions arising from the dual topology framework are assumed to be negligible but could be taken into account if necessary (see section S2 in the Supplementary Information).

In a second step, we calculate the nonequilibrium work W_{int} associated with exchanging the molecules a and b in the membrane environment. This is accomplished by alchemically

switching off the nonbonded interactions of molecules a and b (outgoing) and switching on the non-bonded interactions of molecules a' and b' (incoming) as a function of λ during the alchemical free energy simulation. In terms of the coupling parameter λ, the outgoing energy $U_{\rm v}(\mathbf{r}^a, \mathbf{r}^b\,\mathbf{R})$ is active when $\lambda = 0$, and the incoming part $U_{\rm v}(\mathbf{r}^{a'}, \mathbf{r}^{b'}\,\mathbf{R})$ is fully active when $\lambda = 1$. The weak distance restraint operating between the outgoing and incoming atoms is unchanged during the transformation. The dual-topology potential energy $U_{\rm int}(\mathbf{r}^a, \mathbf{r}^b, \mathbf{r}^{a'}, \mathbf{r}^{b'}, \mathbf{R}; \lambda) = \lambda U_{\rm v}(\mathbf{r}^{a'}, \mathbf{r}^{b'}, \mathbf{R}) + (1-\lambda)U_{\rm v}(\mathbf{r}^a, \mathbf{r}^b\,\mathbf{R})$ is constructed with the time-dependent coupling parameter $\lambda(t)$ to implement this transformation and calculate the work $W_{\rm int}$ as,

$$W_{\rm int} = \int_0^{t_{\rm int}} \left(\frac{\partial U_{\rm int}}{\partial \lambda}\right) \dot{\lambda}(t)\, dt \tag{1}$$

The starting coordinates for this simulation are $\mathbf{r}_0^a$, $\mathbf{r}_0^b$, and $\mathbf{R}_0$ for molecules a, b and the environment, respectively. The initial velocities for a, b and the environment are the velocities at the end of the previous MD step. In addition, the coordinates $\mathbf{r}_i^{a'}$, and $\mathbf{r}_i^{b'}$, taken from the end of the previous vacuum simulation are used for molecules a' and b', respectively. The velocities of a' and b' (incoming) are randomly generated at the corresponding temperature. The cartesian coordinates of all the common core atoms (incoming and outgoing) are constrained during attempted lipid exchange. This choice, satisfies microscopic detailed-balance by construction, greatly simplifies the dual topology treatment and helps reduce the statistical noise in the alchemical nonequilibrium work $W_{\rm int}$ for the lipid exchange. The alchemical simulation for the attempted exchange consists in switching off the nonbonded interactions (intramolecular and intermolecular) of the lipids a, b, and switching on the nonbonded interactions (intramolecular and intermolecular) of the lipids copies a' and b'. All internal covalent terms are kept all along (bonds, angles, dihedrals, etc). The constraints fixing the cartesian coordinates of the common core atoms do not contribute to the transformation.

If the attempted exchange is accepted, the coordinates of the interacting molecules a', b' at the end of this simulation are copied onto $\mathbf{x}_1^a$ and $\mathbf{x}_1^b$ to continue the equilibrium dynamics. If the attempted exchange is rejected, we return to the initial coordinates and

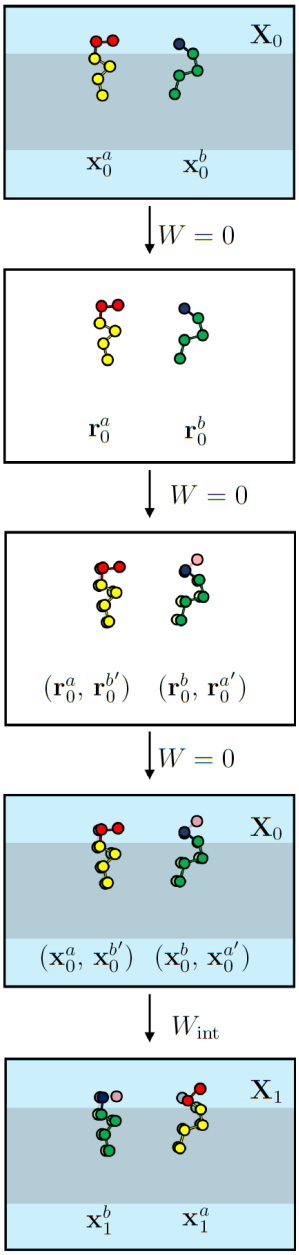


Figure 2: Schematic representation of the lipid exchange algorithm. The work associated with the exchange is evaluated in the step in the membrane. $\mathbf{r}_0^a$ and $\mathbf{r}_0^b$ are the coordinates of the lipids a and b at their initial sites and $\mathbf{r}_0^{a'}$ and $\mathbf{r}_0^{b'}$ at their final sites in vacuum. $\mathbf{x}_0^a$ and $\mathbf{x}_0^b$ and $\mathbf{x}_1^a$ and $\mathbf{x}_1^b$ are the coordinates and velocities of the lipids at their initial (0) and final (1) sites in the membrane. $\mathbf{X}_0$ and $\mathbf{X}_1$ are the coordinates and velocities of the environment at the initial and final states. Red and blue spheres represents the non common atoms of the exchanging lipids, while yellow and green spheres stand for their common part. Lighter colored spheres illustrate decoupled states.

velocities of molecules a, b and of the environment ($\mathbf{x}_0^a$, $\mathbf{x}_0^b$, $\mathbf{X}_0$) to continue the equilibrium dynamics or to do another attempt. A two-ends momentum reversal with a probability of 0.5 is applied before and after slow-growth simulations to symmetrize the probability of

attempted exchanges.[15] Under these conditions, the Metropolis acceptance probability for the attempted exchange,

$$
\begin{aligned}
T^{(\mathrm{a})}(\mathbf{x}_0^a, \mathbf{x}_0^b, \mathbf{X}_0 \rightarrow \mathbf{x}_1^a, \mathbf{x}_1^b, \mathbf{X}_1) = \\
T_{\mathrm{int}}^{(\mathrm{a})}(\mathbf{x}_0^a, \mathbf{x}_0^b, \mathbf{x}_0^{a'}, \mathbf{x}_0^{b'}, \mathbf{X}_0, \lambda_0 \rightarrow \\
\mathbf{x}_1^a, \mathbf{x}_1^b, \mathbf{x}_1^{a'}, \mathbf{x}_1^{b'}, \mathbf{X}_1, \lambda_1) = \\
\min\left[1, e^{-W_{\mathrm{int}}/k_{\mathrm{B}}T}\right]
\end{aligned}
\tag{2}
$$

ensures that the process satisfies microscopic detailed balance, and thus converges toward the equilibrium Boltzmann distribution.[16]

The neMD/MC algorithm for equilibrating phospholipid bilayers is first tested in the case of a binary mixture of the zwitterionic lipid DLPC (1,2-Dilauroyl-sn-glycero-3-phosphocholine) and the negatively charged lipid DLPG (1,2-dilauroyl-sn-glycero-3-phosphoglycerol). These two lipids have the same acyl chains but possess two very different polar heads. By convention, the common atoms encompass all the atoms of the carbon chains and the phosphate groups, whereas the non-common atoms are those of the choline group of DLPC and those of the glycerol groups of DLPG. The procedure for generating the initial positions of the dangling atoms is detailed in section S1 of the Supplemental Information.

A small patch, with 100 DLPG and 100 DLPC per leaflet is created to compare brute-force MD and the neMD/MC algorithm. The lipids are initially sorted (the DLPG in one half of the box, the DLPC in the other half). The initial dimension of this patch is 100 $\times$ 100 $\times$ 93 Å^3 containing around 110,000 atoms. It is hydrated by around 19,700 water molecules with a concentration of NaCl set to 0.1 M neutralizing the overall charge of the system.

Two large patches, with 300 DLPG and 300 DLPC per leaflet, are also studied. To mimic the effect of a membrane protein on spatial distribution of lipids, a transmembrane peptide of 40 amino acids is embedded in the bilayer (Figure 3). It consists of a hydrophobic core of 22 leucine residues capped by 9 positively charged arginine at both ends. The leucine

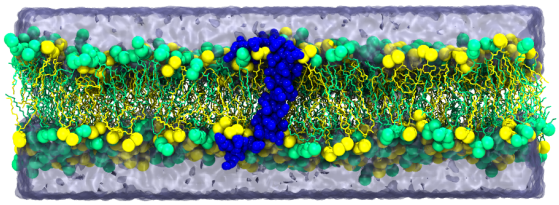


Figure 3: Structure of the homogeneous system. A peptide capped by arginines is embedded in the bilayer. The yellow and green spheres represents the polar heads of DLPG and DLPC, respectively.

segment is shaped as an α-helix and inserted perpendicular to the membrane plane. After a 10 ns equilibration period, its position is restrained during all simulations by means of an harmonic potential ($100\,\mathrm{kcal}\cdot\mathrm{mol}^{-1}\cdot\mathrm{\AA}^{-1}$) applied to the C_α to avoid undesirable movements. The two arginine ends emerge into the solvent on both side of the bilayer. The presence of positively charged residues is intended to favor the enrichment of DLPG against DPLC around the peptide. In one patch, referred to as inhomogeneous, a central circular region of roughly 60 Å in radius contains only DLPC (225 per leaflet), surrounding the transmembrane peptide, while the remaining DPLC are randomly mixed with the DLPG outside the central region. In the second one, referred to as homogeneous, both DLPC and DLPG molecules are randomly positioned in each leaflets. Such a uniform configuration is representative of widely used membrane building procedures.[31] The molecular system is hydrated by 56,000 water molecules with a concentration of NaCl set to 0.1 M neutralizing the overall charge of the system. The size of the resulting simulation box is $190 \times 190 \times 75$ Å^3 containing about 280,000 atoms. The three patches with different initial spatial distributions of DLPC and DLPG are generated using the CHARMM-GUI membrane builder[31](Figure 4).

All simulations are performed in the NPT ensemble using the program NAMD.[32] The nonpolarizable CHARMM36 force field[33] is used for the peptide, ions, and lipids. Water is described with TIP3P.[34] A Langevin thermostat and a Langevin piston are employed to maintain a temperature of 315K and a pressure of 1.0315 bar, respectively.[35] The Particle–Mesh Ewald algorithm is used to handle long–range interactions.[36] Above 12 Å, electrostatic and Van der Waals interactions are truncated at a switching distance of 14 Å. The SHAKE/RATTLE[37,38] algorithm is used to constrain covalent bonds involving hydrogen

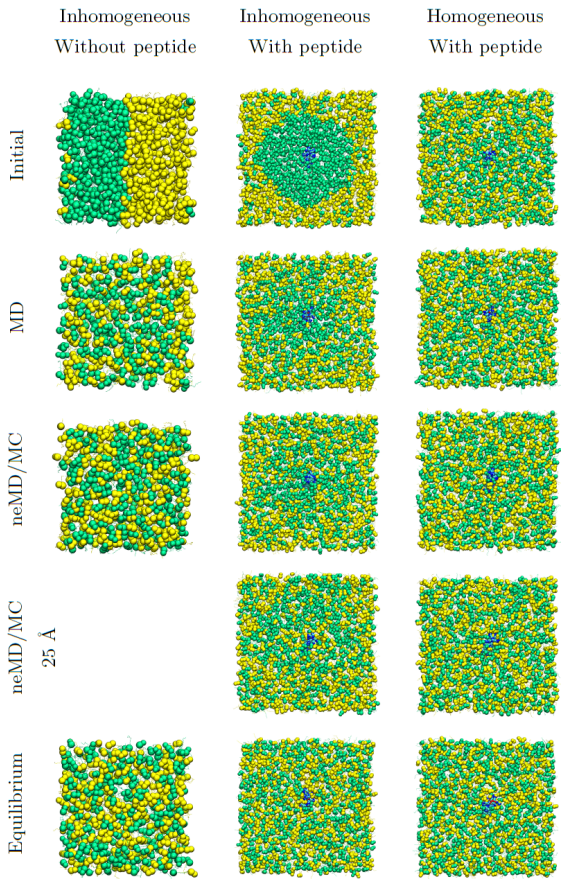


Figure 4: Lipid distribution in the different molecular assays. Bilayer without peptide (first column): The MD distribution is generated after 500 ns, the neMD/MC after 500 successes and the equilibrium spatial distribution after 1 μs of trajectory. Bilayer with the peptide (second and third columns): The MD distribution is generated after 360 ns, the neMD/MC after 360 successes and the equilibrium distribution after 3 μs of trajectory. Yellow and green spheres stand for DLPG and DLPC, respectively. First row: initial distribution, Second row: MD, Third row: neMD/MC, Fourth row: neMD/MC with a selection of at least one lipid within 25 Å of the peptide, fifth row: equilibrated distribution.

atoms and the SETTLE algorithm[39] is utilized for water molecules. In all simulations, the Hydrogen Mass Repartitioning (HMR) scheme[40] is used. All equilibrium MD and neMD/MC simulations were carried out with a time-step of 2 fs.

To serve as standard references, long equilibrium MD simulations exceeding 1 μs are generated for the different molecular assays. All the lipid exchanges are performed using an alchemical dual-topology strategy. No soft-core potential is applied.[41,42] The electrostatics and the van der Waals interactions are decoupled linearly from the system for λ varying from 0 to 1. The neMD/MC simulations rely on the alchemical thermodynamic integration (TI) code previously developed by Radak in the context of constant-pH simulations.[23] The

end goal of the method is to exchange lipids in the most efficient way. Rather than optimizing the acceptance probability, $P_{acc}(\tau)$, of lipid exchanges by increasing the sampling time, τ, of the alchemical transformation, we sought to maximize the exchange rate, defined as $k(\tau) = P_{acc}(\tau)/\tau$.[17] Evolution of $P_{acc}(\tau)$ and $k(\tau)$ as a function of the sampling time are reported in Figure 5 for DPLC/DLPG exchanges in a small inhomogeneous lipid patch. Whereas the acceptance probability increases continuously with τ, the optimal exchange

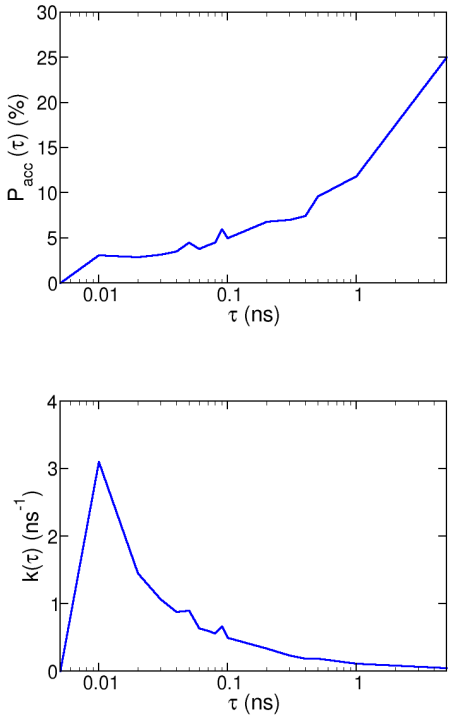


Figure 5: Time evolution of $P_{acc}(\tau)$, the lipid exchange acceptance probability (top) and $k(\tau)$, the lipid exchange rate (bottom)

rate is obtained for an alchemical transformation performed over 10 ps, the corresponding acceptance probability being 3 %. To improve the computational efficiency of the algorithm and maximize the computational time spent for the lipid exchanges, we limited the equilibrium MD to 100 ps, either when an exchange has been accepted or if 33 attempted exchanges have been successively rejected, ensuring that the exchange rate is on average about 1 lipid exchange per nanosecond. The length of the equlibrium MD is independent of the neMD switches and could be made longer if so desired. In practice, 30 and 40 % of

the equilibrium MDs are performed after a successful exchange, whereas the others follow a series of 33 rejected exchanges. For each proposed exchange, we first randomly select one of the two leaflets, then pick at random a couple of DPLG and DPLC lipids. One may note that alchemical simulations often carry some computational overhead relative to straight MD. To minimize the computational cost, we used the NAMD option with direct interactions "alchDecouple off" requiring the smallest number of PME grid calculations per step. The performance of neMD alchemical simulations compared to equilibrium MD decreases by 20% on average.

Two different exchange protocols are tested with the neMD/MC simulations (Lines 3 and 4 in Figure 4). In the first one, lipids are randomly selected from the entire membrane leaflet. In the second one, the choice of lipids is no longer completely random to increase the lipid exchanges attempts near the peptide to enhance the configurational sampling in the region of interest. The membrane is formally divided into an inner region, corresponding to a distance equal or smaller than 25 Å from the transmembrane helix, and an outer region, corresponding to a distance equal or greater than 25 Å from the helix. All attempted exchanges must always include at least one lipid from the inner region (2 inner or 1 inner and 1 outer). No exchanges between lipids in the outer region far from the peptide are attempted. When the attempted exchange involves lipids lying in the inner and outer region, the Metropolis criteria must be modified as,

$$k_{n_a,n_b \to n_a-1,n_b+1} = \left(\frac{n_a(N_b - n_b)}{(N_a - n_a + 1)(n_b + 1) + n_a(N_b - n_b)} \right) \tag{3}$$

where a represents the lipid initially in the inner region (DLPC or DLPG) and b represents the lipid initially in the outer region (DLPG or DLPC). The development leading to eq 3 is described in section S3 of the Supplemental Information. A pseudo code of the neMD/MC algorithm is provided in section S4 of the Supplemental Information.

Let us now describe the main results obtained from the neMD/MC method. A histogram of the alchemical work calculated from neMD/MC trajectories for the small lipid patch assay are shown in Figure 6. Most of lipid exchanges give rise essentially to positive works distributed around a central value of 40 kcal/mol. However, a tiny proportion of exchanges result in works that are negative or weak enough to pass the Metropolis test. It results in a lipid exchange probability of about 3 % corresponding to one exchange every nanosecond, in line with the data reported in Figure 5. The histograms for the other molecular assays are all similar to that plotted in Figure 6 (data not shown).

Although the conformation of the incoming lipid is by design very similar to that of the outgoing lipid, the alchemical transformation may sometimes lead to an end-point catastrophe due to an overlap between the incoming lipid atoms and the environment. Such events occur at the first time step of the alchemical transformation, leading automatically to an exchange rejection without any sampling. Alchemical transformations carried out over longer sampling times show that for this type of conformation the exchange work is always high, resulting in the systematic reject of lipid swapping (data not shown here). It is noteworthy that such rejects, which occur before sampling, do not affect the number of lipid exchange accepted per time unit. We have therefore not taken their number into account when estimating the overall exchange rate.

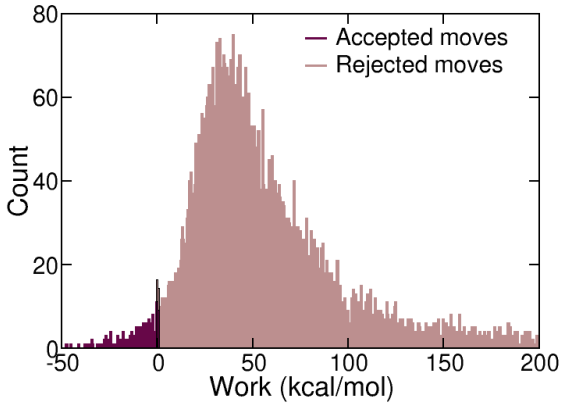


Figure 6: Histograms of the works associated with every exchange attempts in the small lipid patch assay. Brown and indigo stand for rejected (6814) and accepted (200) exchanges, respectively.

To assess the validity of the neMD/MC algorithm, we first sampled the mixing of a DLPC/DLPG mixture starting from an inhomogeneous configuration in which the two lipid components are separated along the x-axis in two adjacent reservoirs. Figure 7 compares the

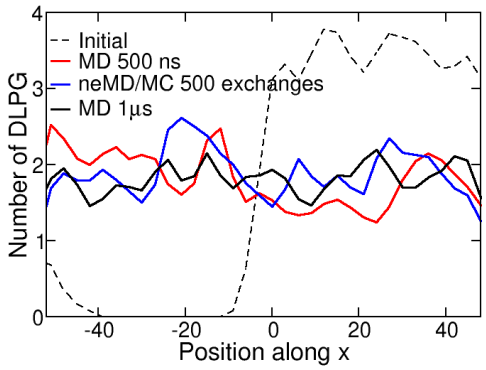


Figure 7: Spatial distribution of DLPG along the x-axis in the small bilayer assay starting from a fully sorted lipid distribution. Initial distribution (dotted line). Distribution after 500 ns of equilibrium MD (red line). Distribution after 500 successful exchanges with the neMD/MC method (sampling time equivalent to 500 ns) (blue line). Distribution after 1 μs of equilibrium MD (black line).

spatial distribution of DLPG obtained after 500 successful lipid exchanges to that observed after 500 ns and 1 μs of brute-force equilibrium MDs. The results indicate that in all three cases, the DLPG have diffused throughout the entire bilayer and their distribution is fairly homogeneous. The detailed procedure for computing the distribution of DLPG is provided in section S4 of the Supplemental Information. To better assess the homogenisation of the lipids, Figure 8 shows the structure factor of the lipids (details are provided in Supplementary Informations). It is observed that after 500 ns or 500 exchanges with neMD/MC, we obtain a homogeneous distribution of the lipids that is close to the result after 1 μs.

Simulation of membrane systems requires not only a correct description of the lipid mixing process, but also accurate sampling of the specific association of lipids with membrane proteins. We therefore carried out a series of neMD/MC and equilibrium molecular dynamics simulations for DLPC/DLPC bilayers embedding a transmembrane helix capped with positively charged residues. Figure 9 depicts the pair radial distribution functions[43] (RDF) between DLPG headgroups and the first arginine linked to the the hydrophobic helix in

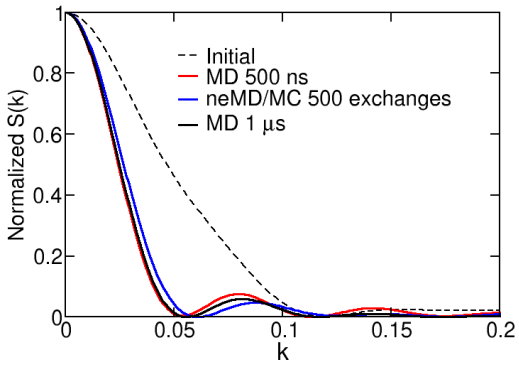


Figure 8: Structure factor of the distribution of the DLPG along the x-axis in the small bilayer assay starting from a fully sorted lipid distribution. Initial distribution (dotted line). Distribution after 500 ns of equilibrium MD (red line). Distribution after 500 successful exchanges with the neMD/MC method (sampling time equivalent to 500 ns) (blue line). Distribution after 1 μs of equilibrium MD (black line).

the two-dimensional membrane plane generated from a neMD/MC simulation with 360 lipid exchanges together with those monitored after 360 ns and 3 μs of equilibrium MD for the different lipid/peptide assays described in Figure 4. Their corresponding integrals are provided in Supplemental Figure S2 of the Supplemental Information. RDFs obtained at different sampling times for equilibrium MD and at different lipid exchange number for neMD/MC as well as the detail of their calculation are provided in sections S6 and S7 (Supplemental Figures S3, S4 and S5). The RDF generated from an initial DLPC/DLPG random mixture built with CHARMM-GUI[31] show that, by construction, the arginine neighbourhood is already slightly enriched in negatively charged DLPG. The latter gradually increases to reach an equilibrium illustrated by the RDF at 3 μs (see Section S6 in Supplemental Information). The RDF obtained after 360 lipid exchanges demonstrates the capacity of the neMD/MC algorithm to generate configurations in line with those sampled after a few μs of equilibrium MD. Biasing the selection of lipids to be exchanged to a region close to peptide does not fundamentally influence the sampling, indicating that for this assay, the initial lipid distribution generated by CHARM-GUI is suitable. For the innhomogenous lipid assay, all the DPLGs are initially distributed at the periphery of the lipid patch at distances greater that 50 Å from the transmembrane peptide. The RDFs reported in Figure 9) show that in

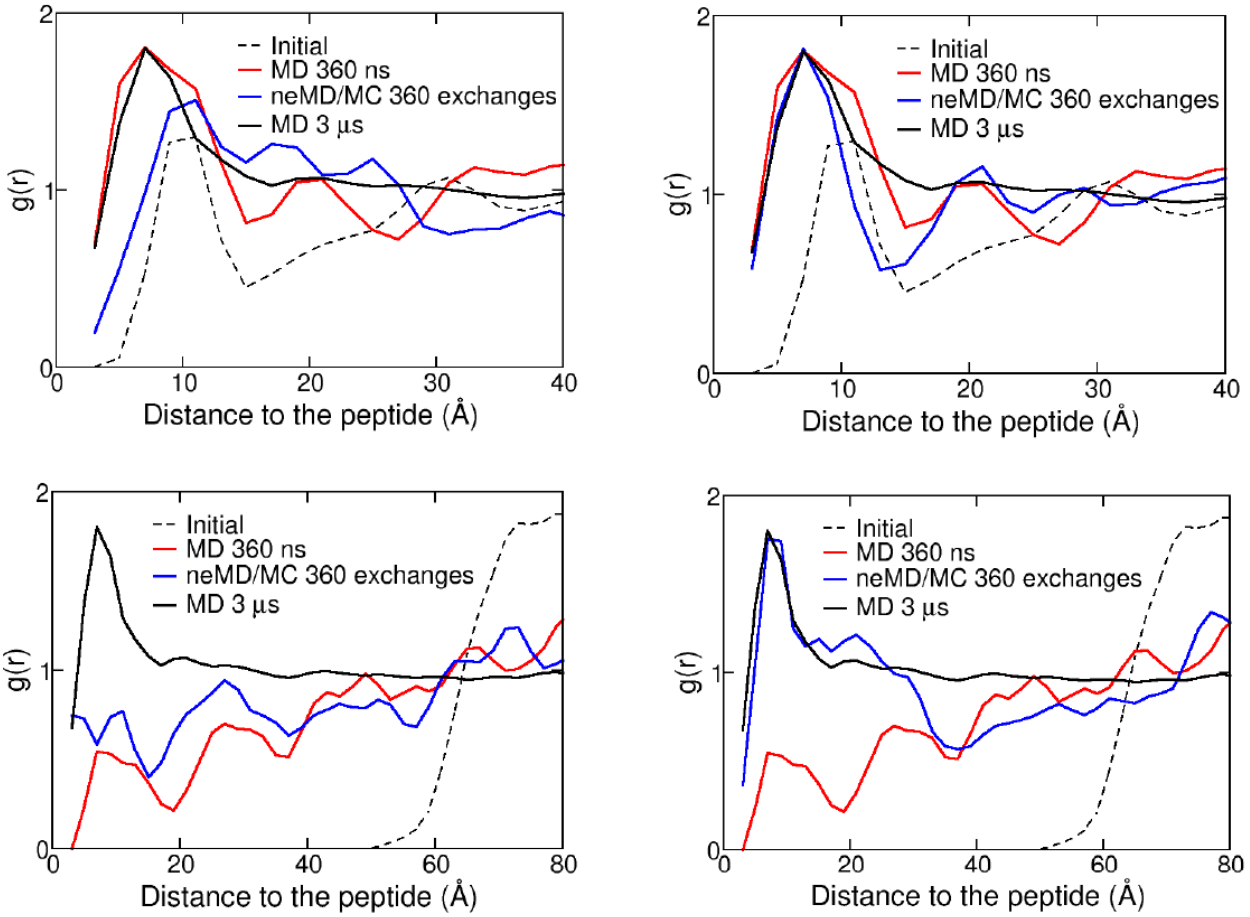


Figure 9: Pair radial distribution functions (g(r)) between the phosphorous of DLPG and the alpha carbon of the peptide arginine starting from a homogeneous (top row) and an inhomogeneous (bottom row) lipid distribution. The homogeneous distribution is shown on a smaller scale (only 40 Å around the peptide), as it is around one by construction far from the peptide. For the neMD/MC simulations, exchanged lipids are selected randomly over the entire system (left column) or at a distance lower than 25 Å from the peptide (right column). Initial distribution (dotted line). Distribution after 360 ns of MD (red line). Distribution after 360 successes with the neMD/MC method (sampling time equivalent to 360 ns) (blue line). Distribution after 3 μs of MD (black line).

360 ns of equilibrium MD the lipids have barely diffused to the central peptide, leaving the system far from the equilibrium observed after several microseconds of MD simulation. In contrast, a neMD/MC trajectory of 360 lipid exchanges (sampling equivalent to 360 ns of MD at equilibrium) generates configurations in which a few DLPGs are bound to peptide arginines. However, the system is still far from the equilibrium observed after 3 μs of simulation, indicating that for large systems the neMD/MC hybrid approach does not overtake equilibrium MD for lipid sampling. When choosing the lipids to be exchanged in a region

close to the peptide, it is clear that the neMD/MC approach can, at least in this region, generate patterns close to those sampled after 3 μs of MD (see Figure 9 and Section S6 in Supplemental Information).

Simple considerations are helpful to clarify the conditions where the neMD/MC lipid swapping algorithm is expected to be the most useful compared to unbiased MD simulations. The most difficult situation to sample corresponds to a protein in membrane comprising a mixture in which one type of lipid is very dilute. In this case, it can take a very long time to recruit the dilute lipid near the protein and allow its overall spatial distribution to relax. Treating the problem the limit of a simple bimolecular association and dissociation process, the global relaxation time is expected to go as $\tau^* \propto 1/D(C + K_\mathrm{d})$, where C is the concentration per area of the dilute lipid, and K_d is the equilibrium dissociation constant of the lipid with the protein. This expression is obtained as the global relaxation time of a two-state system $1/(k_f + k_b)$ where forward rate is $k_f = k_\mathrm{ass}C$ and the backward rate is $k_b = k_\mathrm{ass}K_\mathrm{d}$, with the assumption that the bimolecular association rate k_ass is proportional to the diffusion coefficient D. The relaxation time gets slower at low concentration and small diffusion coefficient. In unbiased MD simulations, the lateral diffusion coefficient D_L of a lipid molecule in the plane of the bilayer is on the order of 1.0 Å^2/ns. However, in the neMD/MC algorithm, the effective lateral diffusion process is completely artificial. Any given lipid could exchange with a uniform probability anywhere in the simulation box of length L. For each accepted move, the mean-square displacement along one axis increases by $L^2/12$. It follows that the effective diffusion coefficient arising from the lipid exchanges follows the relation, $2D_\mathrm{ex}\tau = N_\tau(L^2/12)$, where N_τ is the average neMD/MC moves accepted using a nonequilibrium switching time τ (here, $N_\tau = 1$ when $\tau = 1$ ns). Accordingly, the global relaxation time of the spatial distribution of the dilute lipid around the membrane protein with the neMD/MC algorithm goes as, $\tau^*_\mathrm{ex} = \tau^*_\mathrm{MD}\,(D_\mathrm{L}/D_\mathrm{ex})$. For example, in the case of a 100×100 Å^2 membrane patch in a typical simulation system comprising a membrane protein the effective neMD/MC diffusion coefficient is about 417 Å^2/ns suggesting that, compared to

unbiased MD, the the spatial distribution of a dilute lipid around the membrane protein could relax almost 500 times faster with the neMD/MC algorithm. The accelerated relaxation is expected to be particularly advantageous at low concentration C. It is important to note, however, that this argument pertains only to the translational diffusion. A complete sampling of a membrane also requires sampling the internal configuration of the lipid chains, which occurs on a timescale of about 20 ns.

The hybrid neMD/MD algorithm for exchanging different molecules with a common core is expected to be of practical value as long as their size and shape are sufficiently similar. Future work will seek to expand the algorithm to sample systems comprising mixtures of multiple lipids that differ from their carbon tails,[28] as well as heterogeneous systems giving rise to the formation of liquid-disordered and liquid-ordered nanodomains.[44]

Supporting Information Available

The Supplemental Information includes: Section S1 lipid structures and Supplemental Figure S1. Section S2 Supplemental theoretical developments for configurational coupling in dual topology. Section S3 Supplemental theoretical developments for attempted exchanged between the inner and outer regions. Section S4. Pseudo-code for the neMD/MC algorithm. Section S5. Spatial distribution of DLPG along the x-axis. Section S6. Construction of the 2D radial distribution function. Section S7 Supplemental Figures S2, S3, S4, S5.

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TOC Graphic

